LUKHANYO FLORIS KALASHE

Full-Stack Engineer · Data Engineer · AI Engineer

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PROFESSIONAL SUMMARY

Honours candidate in Computer Science & Informatics with hands-on experience across full-stack engineering, data engineering, and applied NLP. Demonstrated ability to design and ship production-ready systems, from secure ASP.NET Core & Angular SPAs deployed on Azure and Render, to dimensional data warehouse pipelines built in SSIS, to Retrieval-Augmented Generation (RAG) systems developed in Python and LangChain. Proven expertise in developing interactive business intelligence dashboards and optimizing semantic data models in Power BI. Strong foundation in clean architecture, cloud deployment, and translating complex business requirements into maintainable, scalable technical solutions.

EDUCATION

BSc Honours — Computer Science & Informatics *Jan 2026 – Present*

University of the Free State | JSE Bursary Recipient (2026)

- Core modules: Natural Language Processing, Advanced Internet Programming, Data Warehousing
- Focus areas: Transformer models (BERT/GPT), RAG systems, ASP.NET Core & Angular SPAs, SSIS ETL pipelines, dimensional modelling

BSc Information Technology — Computer Science & Business Management *Jan 2023 – Nov 2025*

University of the Free State | Graduated with Distinction

- Relevant coursework: Data Structures & Advanced Programming, Systems Analysis & Design, Databases & DBMS, Internet Programming, Computer Networks, Applied Statistics & Data Analytics

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, Python, SQL, R, HTML, CSS, SAS

Frontend: Angular, TailwindCSS, DaisyUI, Bootstrap, jQuery

Backend & Cloud: ASP.NET Core Web API, Entity Framework Core, SignalR, Azure, Render, REST APIs, JWT, CI/CD, Git

Data Engineering: Power BI, DAX, SQL Server, SSIS (ETL), MySQL, Star Schema / Dimensional Modelling, Slowly Changing Dimensions (SCD)

AI & NLP: Python, LangChain, Hugging Face Transformers (BERT/GPT), spaCy, NLTK, scikit-learn, Pandas, Semantic Search, Prompt Engineering

Architecture: MVC, Unit of Work, Repository Pattern, RBAC, Agile, UML, Cloudinary, Postman

TECHNICAL PROJECTS

DatingApp & Course Tracker - Full-Stack Projects *2026*

- Architected and deployed two complete Single Page Applications using ASP.NET Core Web API (C#) as the backend and Angular (TypeScript) as the frontend, applying every major full-stack concept end-to-end.
- Implemented real-time online presence tracking and live user messaging using SignalR; secured both applications with ASP.NET Identity, JWT tokenisation, refresh tokens, and Role-Based Access Control (RBAC).
- Built the data access layer with Entity Framework Core — explicit many-to-many relationship mapping, Repository and Unit of Work patterns; integrated Cloudinary for scalable photo storage.
- Implemented reactive forms, custom validators, pagination, sorting, filtering, client-side caching, and Angular route guards; styled interfaces with TailwindCSS and DaisyUI.
- Configured CI/CD pipelines and published production builds to Azure and Render via Git.

RAG System – Natural Language Processing (Honours) *2026*

- Engineered a Retrieval-Augmented Generation (RAG) system to automate the analysis and querying of technical software engineering course documents.
- Implemented dense semantic retrieval utilizing FAISS and document embeddings to ensure high-accuracy context fetching.
- Designed a hybrid search approach incorporating TF-IDF to significantly improve retrieval precision for highly technical software engineering terminology.
- Conducted bias and error analysis using LIME; applied advanced prompting and in-context learning techniques to improve system output quality.
- Co-authored an academic research paper documenting methodology and findings; delivered a formal departmental presentation of the system.
- Leveraged advanced Natural Language Processing (NLP) libraries, including spaCy and Hugging Face Transformers, to build a robust, intelligent question-answering pipeline.

Enterprise Data Warehouse & BI Analytics Platform (RooiTea) Honours Data Warehousing Module
2026

- Designed a multi-subject data warehouse for a rooibos tea distribution company, covering production, inventory, and international sales with historical exchange-rate support.
- Compiled a detailed information package: identified and categorised all measures (monetary, counts, time periods), dimensions, hierarchies, and roll-up structures (Farm → Region → Province → Country; Day → Month → Quarter → Financial Year, and more).
- Produced a fully normalised star schema dimensional model with three fact tables (FactProduction, FactInventory and FactSales) and supporting dimensions including DimDate, DimCurrency, DimProduct, DimCustomer, and DimBatch.
- Developed three SSIS packages (MasterPackage, LoadDimensions, LoadFacts) in Visual Studio 2022: implemented Derived Column, Lookup, Conditional Split, Data Conversion, and Sort transformations; configured SCD handling, error redirects, and flat-file error logging.
- Developed a comprehensive 4-page Power BI reporting suite utilizing custom DAX measures to track global sales footprints, inventory liabilities, and manufacturing yield efficiency.
- Successfully resolved Ambiguous Filtering Paths within the Power BI Star Schema by strategically managing active and inactive relationships to preserve exact filter contexts across bridging dimensions (DimBatch) and time-intelligence tables (DimDate).
- Delivered an oral technical defence of the data infrastructure design to the module team.

EXPERIENCE

Tutor — C# Programming & Web Development *Mar 2024 – Nov 2025*
University of the Free State

- Mentored first-year students in OOP principles, C# programming logic, and introductory web development.
- Delivered clear explanations of complex concepts and provided actionable code reviews aligned with software engineering best practices.
- Evaluated and marked technical assignments, maintaining consistent grading standards across a large cohort.

CERTIFICATIONS & EXTRACURRICULAR

- High-Performance Computing Clusters: GNU/Linux administration, VLANs, and secure communication in distributed systems.
- IT Student Association: Active participant in hackathons, technical workshops, and campus technology events.

Transcript and references available upon request.